

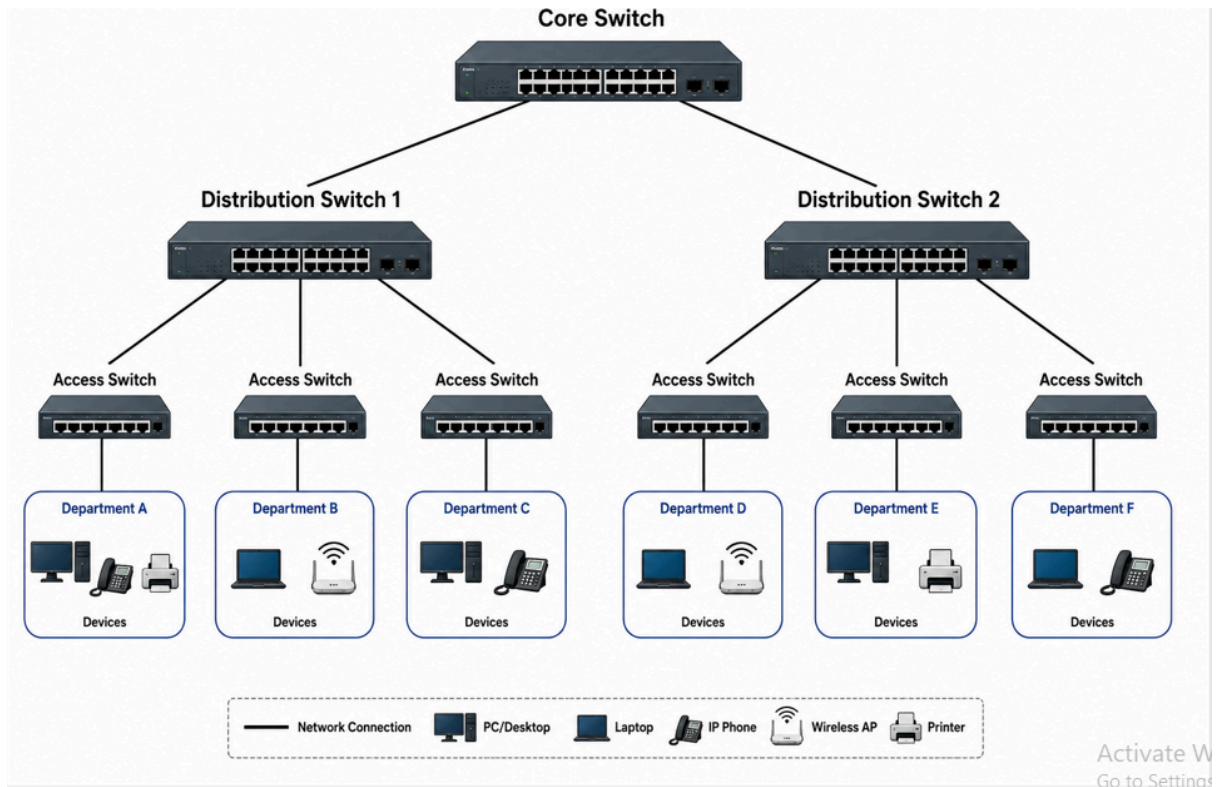
Client Design Review-Tlhogo Roofing & Construction

1.REQUIREMENTS

Requirement	explanation	Why it matters
Connectivity	All departments must have good internet access and service and internal servers.	It enables communication and access to shared resources.
File sharing	Server that stores project documents, blueprints, and tenders.	It prevents data loss and allows collaboration.
Email & Internet	The internet must be reliable for email, research, online tenders, and online meetings.	It is essential for daily operations.
Guest Wifi	Visitors and sub-contractors must have their separate wifi.	It keeps the internal network secure while providing access.
Voice(VoIP)	Office staff must have IP phones.	It reduces telephone costs and integrates with the network.
Security	There must be VLAN segmentation, firewall, and access control.	To protect sensitive data and to prevent unauthorised access.
Scalability	The network should allow adding new users in the future, it must not have a fixed number of users and it will change in the future.	It prevents costly redesigns as the company grows.
Reliability	The network should be reliable	To minimise downtime

2. PHYSICAL TOPOLOGY

I will use the Hierarchical star(extended star topology) because with it, it is easy to add new switches and devices. It is more organised as it has a clear separation of core, distribution, and access layers. It can add backup links without redesigning and it mirrors how enterprise networks are built.



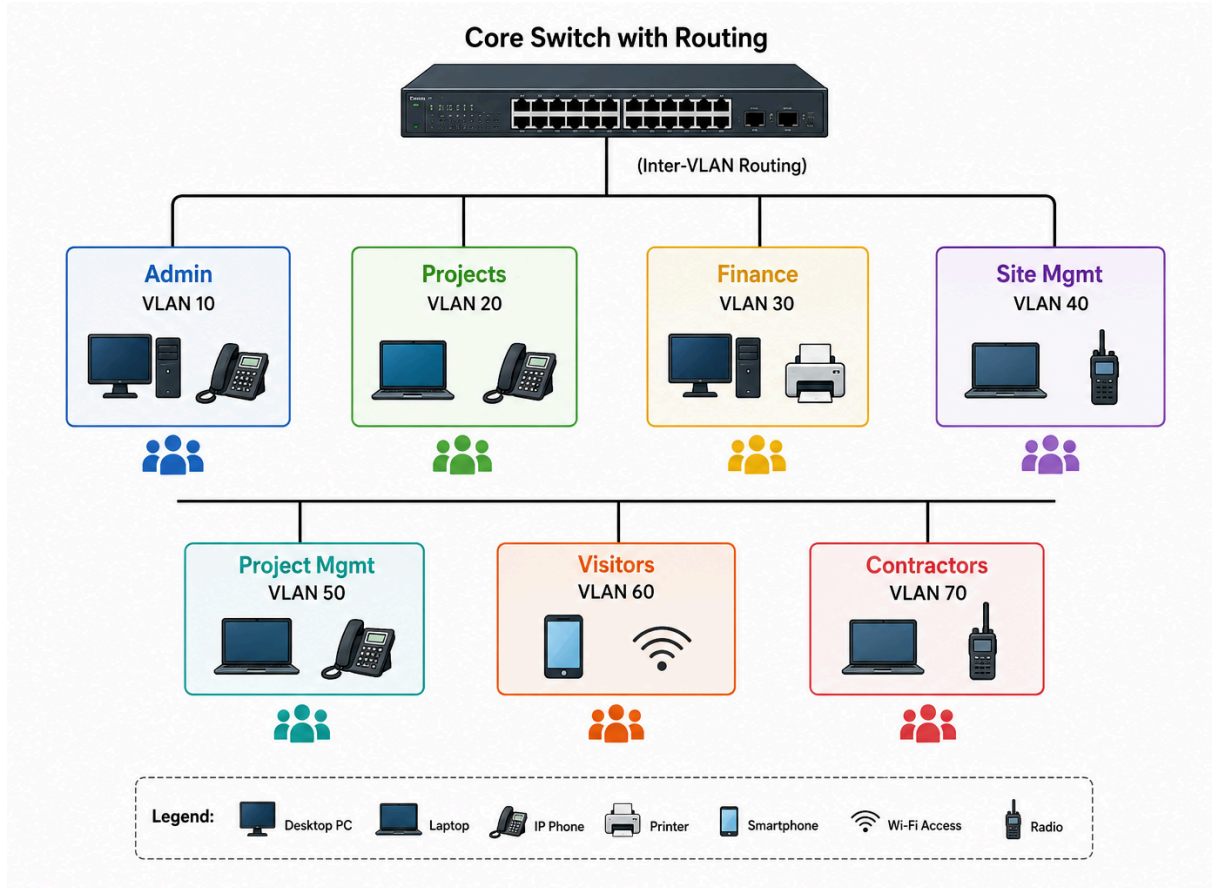
3. Logical topology

The topology to be used is the VLAN Segmentation because it has high security because it isolates departments, it reduces broadcast traffic and it is flexible, meaning that devices can be moved without changing IP addresses. With it, it is easier to add new departments or users.

VLANs and Subnets

VLAN	Name	Subnet	Purpose
10	Admin	192.168.10.0/24	Management and administrative staff
20	Projects	192.168.20.0/24	Project managers and estimators
30	Finance	192.168.30.0/24	Financial and payroll data
40	Site Management	192.168.40.0/24	Site supervisors and coordinators
50	Contractors	192.168.50.0/24	External contractors on-site
60	Project Management	192.168.60.0/24	High-level project oversight
70	Visitors	192.168.70.0/24	Visitors and subcontractors (no

			internal access)
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4. IP Addressing Plan

VLAN	Name	Network Address	Subnet Mask	Gateway	DHCP Range
10	Admin	192.168.10.0	255.255.255.0	192.168.10.1	10.10 – 10.50
20	Projects	192.168.20.0	255.255.255.0	192.168.10.1	20.10-20.50
30	Finance	192.168.30.0	255.255.255.0	192.168.10.1	30.10-30.30
40	Site Management	192.168.40.0	255.255.255.0	192.168.10.1	40.10-40.20
50	Contractors	192.168.50.0	255.255.255.0	192.168.10.1	50.10-50.20
60	Project	192.168.60.0	255.255.255.0	192.168.10.1	60.10-60.20

	Management		0		
70	Visitors	192.168.70.0	255.255.255.0	192.168.10.1	70.100-70.200

5. Resilience and Redundancy

To ensure reliable internet access and minimise downtime, the network will use two routers in a high-availability (HA) configuration. Both routers will be connected to separate WAN links to provide load balancing and failover.

Load balancing distributes traffic across both connections, preventing slowdowns when many users are online.

Failover ensures that if one connection fails, the other automatically takes over.

VLANs will be used to isolate departments, and Quality of Service (QoS) will prioritise critical traffic such as VoIP and video conferencing.